MTH 377/577 Convex Optimization
Problem set

1. Let $f : R^n \to R$ be a convex, continuously differentiable function. Consider a compact subset $D \subset R$. Does f have a maxima/minima in D? Ans. Yes [Weierstrass Theorem].

2. Using KKT conditions, Solve the following:

$$\text{minimize } x_1^2 + x_2^2 - 4x_1 - 4x_2$$

$$s.t.$$

$$x_1^2 \leq x_2$$

$$x_1 + x_2 \leq 2$$

   Ans. (1,1) is the global minimizer. [Find vectors of first derivatives for $f_0, f_i's$ and apply KKT as demonstrated in lectures 11-12.]

3. Find the dual of the cone generated by the vectors in matrix A:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

   Ans. The vectors in A generate the cone $K = \{\theta x | x \in \{(1,0,0),(0,1,0),(0,0,1)\}, \theta \geq 0\}$. Note that $K = \{(x_1, x_2, x_3) \in R^3 | x_i \geq 0 \forall i = 1,2,3\}$. This cone is self dual- its dual is R^3_+, $\{(y_1, y_2, y_3) \in R^3 | y_i \geq 0 \forall i = 1,2,3\}$. [Show formally using definition of dual: $K^* = \{y \in R^n | y \cdot x \geq 0 \forall x \in K\}$].

4. Consider two hyperplanes (h_1, β_1) and (h_2, β_2) such that their halfspaces have a non-empty intersection. Is the intersection of their halfspaces a convex set? Ans. Yes. Half spaces are always convex and the intersection of two convex sets is convex. You may also show using standard definition of convex sets.

5. Write the following optimization problem in standard form and solve it using KKT conditions.

$$\begin{gathered} \textit{maximize} \quad f(x,y) = 2x + 3y \\ s.t. \\ x^2 + y^2 \leq 1 \end{gathered}$$

Ans. Note that the inequality constraint needs to be written as $f_i(x, y) \geq 0$ for standard form. Therefore, write $-x_1^2 - x_2^2 + 1 \geq 0$. Solve on your own.